Research Article

Misaligned Stop Code Errors: A Contextual Analysis of How Error Message Conventions are Perceived

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Abstract – Background: In July of 2025, Microsoft had released the (at the time) newest version of the Windows 11 Operating System (OS), 24H2. With this update, came a foundational change to how blue screen errors are reported to the user. A tonal shift was made, changing the blue screen for a black screen and removing the graphics, leaving only stop error codes and helper text. This study observes the effectiveness of this new update in relation to stop code error messages and the level of comprehensibility that allow users of all levels to mitigate their critical system failure. ***Literature Review:*** There is a fundamental flaw in the way stop code error messages are designed where general users and IT professionals are unable to gauge if the error message "is successful in affecting the correct user behavior"; or in other words, neither audience is provided with enough information to "fully understand the warning" and thus label these messages as "an overall failure of the warning". ***Research Question(s):*** *1. Does the design of Windows 11 24H2 stop code error messages provide general users and IT professionals with enough information to identify the type of issue and actionable steps to mitigate the issue? 2. Does this new design serve as an effective means of usability to the indented audiences of general users and IT professionals?* ***Methodology:*** *62 user posts were recorded evenly between the r/WindowsHelp and the Windows Eleven Forum that had experienced a stop code error. Each post was recorded based on the comprehensibility of the stop code error presented and the ability for users or IT professionals to successfully mitigate the critical system failure.* ***Results and Discussion:*** *It was concluded that although Microsoft was cognizant of the limitations from the previous error message design standard, there is still a lack of issue identification and mitigation direction when reported on by the two audiences. 69% of reported stop error codes were unable to be fully understood and 65% of reported cases were not successfully resolved. As coined within this study, there is a clear misalignment with who these error messages are positioned for. The collected data showed that when presented with the newly designed stop code error messages, both IT professionals and general users struggled to correctly identify the issue plaguing the impacted device.* ***Conclusion:*** Windows stop code error reporting is not effectively providing our users or IT professionals with enough with enough information to satisfy either of their needs. This is creating a cycle of complains that lead to more devices being impacted, more customer user frustration, and more IT help backlogs. As it currently stands, the Windows OS does not seem to be able to effectively support any of the types of users when presenting a stop code error. Instead, falls short with broad and inconsistent error reports that may not be indicative of the issue at hand.

Index Terms *- Stop code error, BSOD, Ad-hoc, Operating system, Drivers*

I. Background

SINCE the introduction to the Windows Operating System (OS) back in 1985, the standard convention of computational error message reporting has, at most, shown slight iterations between versions. However, the first major change to this convention came with the release of Windows 11 24H2 in July 2025 where the blue screen that was present for 40 years had been replaced by a black screen. The typical text and automated error message still remain noting that the system encountered a critical error, but this change presents an opportunity to observe the effectiveness not only in the actual message being reported, but the perception in how this black screen parallels the critical system failure.

When these stop code errors occur, the two groups of individuals tasked to handle them are either a general user or an IT professional. In both cases, the message contents are essentially the same. How they are interpreted are two vastly different outlooks. For a general user, there should be a level of understanding on what the device is attempting to tell them. There should be information that allows them to learn from the issue in hopes to either make an initial mitigation step or prevent it happening in the future. For an IT professional, there should be a level of consistency and clarity for immediate mitigation steps to be made without blindly troubleshooting for a solution.

What is shocking is that after all the updates made to the Windows OS and effort to stay ahead of emerging competitors such as MacOS or Linux-based OS, these messages have essentially stayed the same. Besides this recent color change, the main components of these messages have not changed and continues to leave the audiences involved stuck in limbo during an overwhelming and anxiety-inducing event.

In this study, I observe these instances where general users and IT professionals attempt to mitigate issues stemming from the newly designed stop code error messages and how this convention is perceived from both parties involved. I aim to support my argument that the design of stop code

error messages is an instance of a mismatched intended audience. While much of the literature I have collected points to error messages needing to fully embrace user experience where the design should actively change alongside the generations of devices, what I have observed is that the perception from general users show that these error messages have not truly changed. There is a resistance against these messages as they have failed to be transitioned to accommodate modern user reporting (Shneiderman 1982). The data I have collected points to the design of stop code error messages lacks clarity in the actual issue the device is experiencing, leaving general users and IT professionals to both mitigate the issue without any indication of how to prevent this issue in the future.

II. Literature Review

From a fundamental standpoint, computational error messages vary in depth and severity ranging from a common "404 Not Found" message to the detrimental Blue Screen of Death error (BSOD). Each message is designed to emphasize the critical failure and utilize the remaining computing function that is left to display this report on screen. Effective error message design follows the strategies presented by Ben Shneiderman in increasing attention, establishing quality control, developing guidelines, carrying out acceptance tests, and collecting user performance data while simultaneously emphasize "context-insensitivity" through "logical argumentation and nonanthropomorphic messages" (Shneiderman 1982; Taipalus and Grahn 2023). However, when these elements are ignored, the intended audience of general users and IT professionals are left unarmed to combat the issue, wildly taking actions in the hopes of correcting the problem "without making an actual effort to understand the problem" (Traver 2010).

This chaotic clash is why stop code error messages could be perceived as "paradoxical"; these messages are intended to be used as tools for general users and IT professionals alike to diagnose an issue but are left a minimal sense of direction and forced to learn on their own "how to solve the errors or just end up accepting the error message" (Traver 2010). However, the Windows 11 24H2 update attempts to correct this perceived paradox in changing the convention of the "Blue Screen of Death" to the "Black Screen of Death". Looking at this strictly from a color perspective, the act of changing blue to black fits the unpleasant tone that these messages should be representing. This change stems from a relation between color and cognitive learning as it the value is shown through the emphasized relevant cues (Pett and Wilson 1996). This cue is the negative connotation behind the message. The original blue background was believed to provoke an acuity for the "stimulus materials" (the device's issue) and a sense of "cool, passive, peaceful, and controlled events" (Pett and Wilson 1996). Instead, this was a misaligned tone as the critical nature of these error messages equated to a life-threatening alignment for a user's device, thus the change to a black background to parallel this emotional appeal in an attempt to make the nature of the issue match the image on screen. In practice, this does achieve a sense of proper audience analysis where Microsoft situated the error message "to make it work seamlessly with the affordances of the interface contained in the design so that the meaning contained in the writing is clearly communicated" (Jeyaraj 2023).

Despite the effort to modernize the "Blue Screen of Death", these error messages lack tangible and actionable information that an IT professional requires to effectively mitigate the issue at hand. The overall convention is tailored for general users to recognize their device has hit a fatal error but lacks consistency or comfortability for users to alleviate the anxiety or confusion on their device's status. In these overwhelming instances, it would be more beneficial to general users if they were presented with a clear, simple set of text that "only presented essential information in a logical flow" that also prepares them to "navigate complex problems with greater ease and effectiveness" (Ponnada 2020).

If the user is to be the primary audience, then the error messages should be designed to be "procedure-oriented", or "contain information on how the system actually works" and "sets of operating instructions arranged in logical patterns according to the target audience's goal in using the software" (Rosenbaum and Walters 1986). There should always be an opportunity for users, both professional and general, to engage with a "human-forward design" that enables "more control of their interactions, allowing them to guide the conversation in the appropriate direction" (Ponnada 2020). This part is critical to effective error message design. It is one thing to be a physical warning to users, but it is another to actively engage with them and allow them to have a choice in a response. There is an open dialogue that occurs when this happens between the user and the impacted device, thus, granting them power in the mitigation decision.

If these messages were tailored for IT professionals to be the primary audience, then the messages should be designed to view "software from the inside – from the standpoint of the commands, algorithms, and routines that determine how the program acts upon certain data structure" (Rosenbaum and Walters 1986). Instead, the design of modern stop code error messages achieves neither requirement

for their two audiences. Rosenbaum's sentiment on this is that approaching computational materials such as stop code errors "cannot be validated until the target audiences actually begin using the documentation", the iterations of the Windows Operating System should have provided enough "user testing" to realize the middle ground necessary to address both audiences. Both parties obtain the proper information where "novices gain additional expertise, while still addressing the needs of readers who are already experts – even, if possible, to increase the experts' knowledge – without forgetting the existence of hidden audiences" (Rosenbaum and Walters 1986).

III. Methodology

The methodology I used to collect data for this study was a contextual analysis of stop code error reports posted on two popular online forums: r/WindowsHelp and the Windows Eleven forum. Both forums are active communities where general users can reach out when looking to troubleshoot an issue. These can range in severity; a user could ask how to change a simple display setting to wonder why their Operating System won't boot to the home screen. For this study, I examined posts that clearly identified when a stop code error message appeared.

The first forum used was the Reddit thread r/WindowsHelp. This specific Reddit thread allows Windows users to create help posts for others in the community to either identify the specific code presented on their device or offer instructions on how to mitigate the issue at hand. The members of this community vary in their computing knowledge but are centered in general users. Pairing with this forum, I also collected data from the Windows Eleven Forum. This site presented similar material where general users were able to reach out on more information regarding the error code they were experiencing; however, the Windows Eleven Forum had dedicated IT professionals respond to help mitigate the reported issues.

With each forum, I analyzed user posts over the course of a year span (April 2025 to April 2026). For each post, I took note of the following elements:

- Error code reported – The specific line of code the user reported on experiencing.
- Level of comprehensibility – If the user was seeking complete troubleshooting help or understood the issue and looking for actions to mitigate.
- Impacted device – What device the user reported the error code on.
- If an IT professional was involved
- Suggested mitigation – What actions were suggested to attempt to resolve the issue.
- If the issue was resolved
- Issue category – What component of the device required further mitigation.

The above elements were the key indicators on how the error messages were perceived, comprehended, and acted on. For each post, I read through what the reporter had been experiencing and the accompanying responses from either other users or IT professionals.

In total, I recorded data on 62 posts, split evenly between those found on r/WindowsHelp and the Windows Eleven Forum. This was done intentionally based on the 31 total stop code error reports made on the Windows Eleven Forum. I wanted to have an even data set between both forums, so I recorded the most recent 31 posts on r/WindowsHelp to pair with this. Taking data from this calendar year also allowed me to focus on the newly released Windows 11 24H2 version of the stop code error messages. This update marked the first major overhaul to the "Blue Screen of Death" convention where the sad face image and QR code had been removed thus the posts recorded provided perceptions on this new convention. I recorded data from posts that either included an actual image of the error message or a report log from their device. This was my form of quality control to ensure the recorded posts were experiencing a stop code error rather than misreporting the issue.

IV. Results/Discussion

Out of the 62 posts I analyzed, a total of 40 unique error codes were presented to users. However, in a few cases, the unique code was presented differently from one case to another. For example, a user was issued a "0x0000003B" while two others received a "SYSTEM_SERVICE_EXCEPTION" error message. These two messages mean the same thing; both indicate a corrupt or outdated driver. In addition, many of the posts noted an inconsistency with the new color convention where an initial error message would appear with the older blue screen, but a follow up would result in the black screen appearing.

Following this level of comprehensibility, the error messages led to a split in how they were perceived. To collect this information, I observed how the initial post was worded and noted whether the reporter had no sense of direction on what the code meant or what steps to take. If this was the case, I marked this as a troubleshooting-level of comprehensibility. If the reporter had attempted to mitigate this issue and had a general sense of what needed to be done, I marked this as a mitigation-level of comprehensibility. As seen in the below chart, 43 of the 62 posts had a troubleshooting-level of comprehensibility compared to the 19 that held a mitigation-level of comprehensibility:

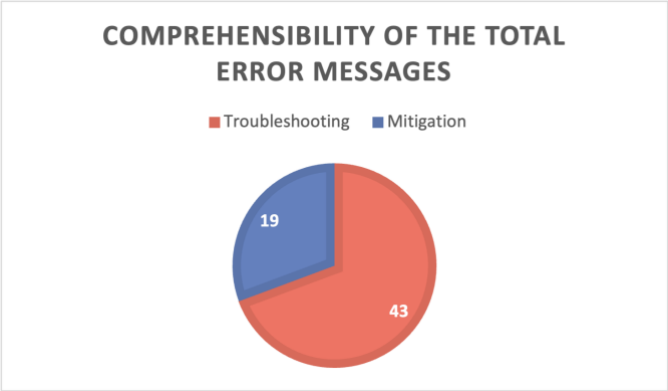


Fig. 1. Pie chart of reported error message comprehensibility.

These posts also held the same number split when looking at reports where a general user sought help from other general users and when an IT professional was sought out. In both instances (31 cases each), 22 held a troubleshooting-level of comprehensibility compared to the nine mitigation-level comprehensibility.

Looking at the actional responses, the posts were split into how the issue was resolved. If the original requestor made note that the error message had been mitigated, I marked this as a resolved issue. If they had mentioned that the suggested actions did not result in a resolution, I marked these as no issue resolution. Finally, if the requestor did not follow up with confirmation or never commented following the initial post, I marked these an unknown resolution.

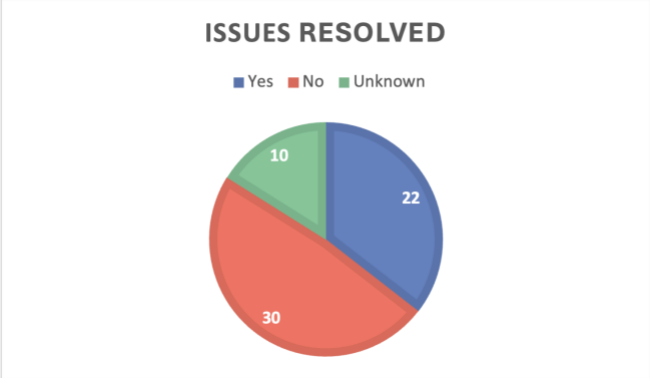


Fig. 2. Pie chart of whether a reported issue was resolved.

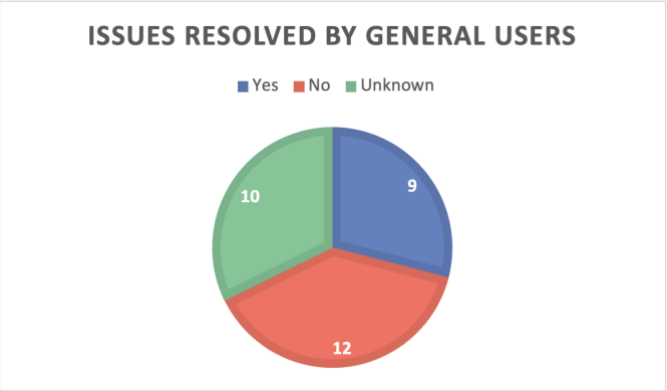


Fig. 3. Pie chart of whether a reported issue was resolved by a general user.

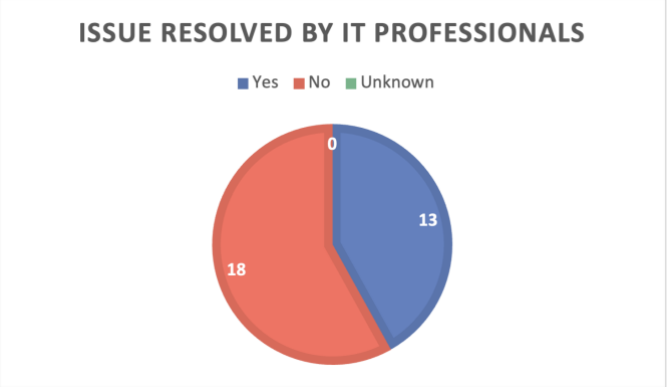


Fig. 4. Pie chart of whether a reported issue was resolved by an IT professional.

As shown in the above graphs, a majority of the reported error messages did not find a sound resolution. Once split between whether a general user or IT professional offered a suggested resolution, the same divide stands with over half the cases with general user advice led to an unknown or no resolution. Even when assisted by an IT professional, 18 reported error message cases were left with no resolution. To pair with this information, I also took note of what action was suggested leading to a potential resolution. In total, there were 14 different mitigation suggestions made between general users and IT professionals across the 62 posts.

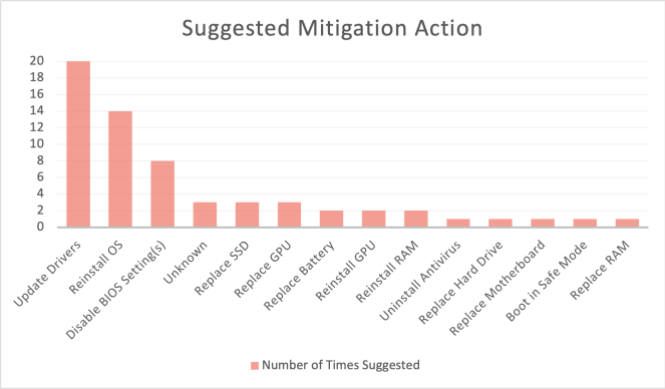


Fig. 5. Bar chart of the mitigation suggestions found in user reports.

As shown in the above chart, the most prevalent suggestion lies with updating the drivers of the impacted device. Despite the expertise in the subject matter, IT professionals regularly were unable to fully resolve the error message at hand (even when provided an image of the stop code error message screen and backlog from the impacted device).

Finally, within the 62 collected posts, 26 of these showed a stop code error occurring on custom built devices such as a dedicated gaming desktop with a mix of installed hardware. This does not directly indicate a theme of what these error messages occur but does add to the validity that these posts came from general users. Broadly speaking, custom built "rigs" are commonly used for gaming purposes that are created solely to withstand extensive graphical and memory loads. This was even furthered by how this was reported in individual posts. For example, when presented with the memory logs on the Windows Eleven forum, initial reporters would include an overview of when the issue occurred. Within [need total] of these posts, the reporter mentioned that they were playing a specific game or whether they were maximizing the output of a specific part (commonly referred to as "overclocking").

Through the collected literature and data analysis, the modern convention of stop code errors shows a limitation on who the error is supposed to be digested by. The Windows Operating System, at times, reports back to these error messages interchangeable from the exact code to the textual identifiers (and even with following the new black design convention). This lack of consistency echoes the foundational strategies mentioned earlier with establishing quality control; the fact that two different error messages can be presented that mean the same thing adds to the level of confusion or incomplete knowledge leading to the general users and IT professionals to lack the necessary comprehensibility in the message (Shneiderman 1982; Taipalus and Grahn 2023).

The error messages simply did not provide enough tangible information for the IT professionals to accurately identify the problem area thus going against the principal Shneiderman presented in increasing attention to message design. Even though they were recently updated, the context of the stop code error messages neglect careful consideration in clarity and consistency leading to ineffective mitigation solutions that ultimately force general users to counterproductive troubleshooting; the reporters work with these ad-hoc actions, thus "committing errors for the sake of receiving hints and suggestions" to an accidental solution (Shneiderman 1982; Taipalus and Grahn 2023). This ties into my assertion that these error messages are designed without proper care to who their intended audience truly is. 69% of the analyzed user posts had no leads or reasonable steps to mitigate the apparent issue at hand. Comparatively, 66% of the analyzed users were not able to effectively resolve their issue. These two percentages show how misaligned the error messages are and provide enough reason to argue that a drastic change needs to occur for these messages to be effectively perceived by general users and IT professionals. Even when an attempt to mitigate the issue was presented, the afflicted area could not be properly identified, leaving both general user and IT professionals to suggest common and broad-ranging actions as shown in the suggested mitigated action. Consistency, clarity, and effectiveness are all omitted from what should be emphasized within the design of these stop code error messages.

While not a main component to this study, I wanted to briefly touch on how my findings on stop code errors fall in line with the Communication Human Information Processing (C-HIP) model. This model "describes the different factors that contribute to warning effectiveness and how information flows and is processed by a human receiver" (Pieterse and Gelderblom 2018). For example, when presented with a stop sign, drivers are able to process the shape, color, and message to understand they need to hit the brakes and stop their vehicle right at that sign. The cognitive process behind the message is clear, understandable, and actionable.

The part lacking with stop code error messages, when following this model, is the last portion of the actionable output. When taking an error message, there should be a reaction from the user. There should be an action to take that responds to what is being processed in the message. However, there is nothing to effectively encode for the user. The only takeaway for both audiences is that something went wrong, and there is a code to look into. What to do with this code is not clear. What the code represents is not clear. It is a clear indication that regardless of the type of user, their necessities are not being met.

As shown through my analysis, there is a fundamental flaw in the way stop code error messages are designed where general users and IT professionals are unable to gauge if the error

message “is successful in affecting the correct user behavior”; or in other words, neither audience is provided with enough information to “fully understand the warning” and thus label these messages as “an overall failure of the warning” (Pieterse and Gelderblom 2018). The errors provide just enough information to gain attention to the user but limit actionable steps to understand the exact issue the device is facing and consequential information to prevent or sustain the device’s ailment.

The critical element here is omitting the ability to connect to what the user needs. The essence of a stop code error message is connecting the user to the device. It is one of the few instances where the device will directly reach out to the user to indicate they are experiencing a problem. In that moment of communication, the provided message should have the “ability to anticipate a user’s needs and deliver them” thus creating “a more holistic experience” between user and device (Ponnada 2020). There is an active relationship that essentially gets ignored through these messages. In a way, this misalignment of audience members can be taken to a third, the device itself. All members must equally be accounted for and what Microsoft has developed negates any of that cross-communication from occurring.

V. Conclusion

The main limitation I had to work around for this study was the lack of data on IT professionals reporting on a stop code error. All the collected data came from general users reporting a specific error code within their personal device. I believe this study could be furthered by gathering data directly from IT professionals; either finding public logs that identify them as the main reporter or from a quantitative survey detailing the perceptions that these professionals have on the design of the stop code errors.

To start this study, I administered a survey to a group of IT professionals. However, they were unable to fully report back due to scheduling conflicts. I feel the data collected from this survey would bolster my argument and cover a wider understanding of stop code error perceptions. Another limitation was deciphering whether an issue was successfully resolved. As shown in the data, 22 of the 62 cases led to an unknown resolution. In these instances, the reporter either never followed up with confirmation of a resolution, or the case was still open at the time of recording.

Another limitation for this study was only focusing on the r/WindowsHelp and Windows Eleven forums. Especially with r/WindowsHelp, there are numerous Reddit communities that focus on this exact topic of discussion. However, the chosen Reddit community was the most active and popular based on their reported contributions (583,000 weekly visitors and 2,000 weekly contributions). Given more time and resources, I would expand this to look into other communities as well as other means of error message reporting. The scope of this study can be taken beyond stop code error messages and operating systems entirely. For example, analyzing the effectiveness of kernel panic errors commonly reported on Mac devices.

This study only focuses on one common iteration of a computation error message. I believe that further research could explore this as a genre further. Error messaging is a key component to how users directly communicate with the software, devices, or programs that they interact with on a daily basis. Whether that comes from attempting to connect to a corporate VPN or unable to join a friend’s online game, these messages constantly appear and require a great deal of rhetorical understand to fully digest what is being transmitted between device and user. This standard practice of error message reporting is what I feel needs to be explored further in the hopes of creating a better user experience during moments of computational error or failure.

Given more time and resources, this study can be furthered to represent a much-needed mark on error message design scholarship. While the discussion here only looks at stop code error messages, there are plenty more that require deeper analysis in order to situate devices to effectively represent their proper audiences. A resolution to this issue would almost be impossible, and changing this convention would require a monumental shift in how the Windows Operating System functions. However, there is an opportunity here to develop a solid answer to this misalignment through a larger study. What I have discovered within this study merely starts this conversation and looks forward to how general users and IT professionals are treated and respected when viewed as intended co-authors.

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